

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03-08-2026	Duration:	60 minutes

Code of Conduct

I Spandana S / 192525160 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction
 - Problem Description

- Scope of the System
 - Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
5. **Identify Assumptions and Constraints**
 - List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
 6. **Validate Requirement Completeness and Consistency**
 - Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential

**SIMATS ENGINEERING
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Department of Computer Science and Engineering

SOFTWARE REQUIREMENT SPECIFICATION (SRS)
Intelligent Smart Port Operations and Cargo Management Platform

CSA1013 — Software Engineering
CO2 Assessment Tool 1 — Scenario-Based Requirement Analysis

Submitted by
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Faculty In-Charge
Latha

Rubric Compliance Summary

This report is organised to directly mirror the five grading criteria of the CO2 Assessment Tool 1 rubric, so that each criterion can be located and evaluated independently.

#	Rubric Criterion	Covered in	Marks
1	Scenario Understanding and Problem Analysis	Section 1	/5
2	Functional and Non-Functional Requirement Identification	Section 2	/5
3	Actor Identification and Use Case Modeling	Section 3	/5
4	Software Requirement Specification (SRS) Documentation	Section 4	/5
5	Assumptions, Constraints, and Requirement Validation	Section 5	/5

1. Scenario Understanding and Problem Analysis

1.1 Business Domain and Industry Context

Modern seaports handle large and continuous volumes of cargo and vessel traffic, which demands tightly coordinated logistics and real-time operational visibility. Technologies such as Artificial Intelligence (AI), the Internet of Things (IoT), Radio Frequency Identification (RFID), Global Positioning System (GPS), and cloud computing are increasingly used to enable intelligent cargo tracking, automated scheduling, and predictive analytics. These technologies collectively improve port productivity, reduce vessel turnaround time, and lower logistics costs, positioning the smart port domain within the broader maritime logistics and supply-chain-technology industry.

1.2 Stakeholders

- Port Authority / Port Operations Management — owns overall port operations and performance targets.
- Cargo Handling and Terminal Staff — perform loading, unloading, and physical movement of cargo.
- Vessel Captains and Shipping Agents / Lines — schedule vessel arrivals, departures, and berthing.

- Warehouse and Yard Managers — coordinate storage, staging, and dispatch of cargo.
- Customs and Port Security Officials — verify documentation and monitor cargo security and compliance.
- Logistics and Freight-Forwarding Companies — track shipments on behalf of clients.
- System Administrators / IT Department — maintain platform configuration, users, and integrations.
- Port Customers / Cargo Owners — expect timely, transparent updates on cargo status.

1.3 Existing Challenges / Problem Description

Port authorities frequently experience operational inefficiencies caused by cargo congestion, manual and paper-based documentation, delayed vessel scheduling, and limited real-time visibility of cargo movement. These issues increase vessel turnaround time, raise logistics costs, and reduce overall supply-chain reliability. There is a need for an intelligent smart port management platform that can monitor cargo movement using IoT devices, optimise berth allocation, predict cargo-handling delays using AI, automate documentation processes, and provide real-time operational dashboards, while enhancing logistics efficiency and supporting secure, transparent port operations.

1.4 System Objectives and Scope

Objectives: monitor cargo movement in real time; optimise berth and vessel scheduling; predict cargo-handling delays using AI; automate logistics documentation; improve warehouse coordination; generate operational performance reports; enhance cargo security; and increase overall port operational efficiency.

In-Scope:

- Real-time monitoring of cargo movement using IoT sensors and RFID tags.
- Berth allocation and vessel scheduling optimisation.
- AI-based prediction of cargo-handling delays.
- Automated generation and processing of logistics documentation.
- Warehouse and yard coordination for inbound and outbound cargo.
- Real-time operational dashboards and performance reporting.
- Cargo security tracking and access control at terminal gates.

Out-of-Scope:

- Physical automation of cranes, cargo-handling machinery, or autonomous vehicles.

- Financial settlement, invoicing, and customs duty calculation systems.
- Crew welfare, payroll, or human-resource management modules.

2. Functional and Non-Functional Requirement Identification

2.1 Functional Requirements

The following functional requirements (FR) describe the observable behaviour the system must provide.

ID	Functional Requirement	Priority
FR-01	The system shall capture and display real-time cargo location and movement data received from IoT/RFID devices.	High
FR-02	The system shall allocate berths to arriving vessels based on vessel size, cargo type, and berth availability.	High
FR-03	The system shall generate and update vessel scheduling timetables and notify relevant stakeholders of changes.	High
FR-04	The system shall use AI/predictive analytics to forecast cargo-handling delays based on historical and live operational data.	High
FR-05	The system shall automate generation, validation, and storage of logistics documents (manifests, bills of lading, gate passes).	High
FR-06	The system shall coordinate warehouse storage allocation and dispatch scheduling for incoming and outgoing cargo.	Medium
FR-07	The system shall provide role-based, real-time operational dashboards summarising cargo, vessel, and berth status.	High
FR-08	The system shall generate periodic operational performance reports (turnaround time, throughput, delay statistics).	Medium
FR-09	The system shall track cargo security status and restrict terminal access to authorised personnel and vehicles.	High
FR-10	The system shall send automated alerts/notifications for schedule changes, predicted delays, and security exceptions.	Medium
FR-11	The system shall allow administrators to manage user accounts, roles, and access permissions.	Medium

ID	Functional Requirement	Priority
FR-12	The system shall log all cargo and documentation transactions for audit and traceability.	Medium

2.2 Non-Functional Requirements

Category	Requirement
Performance	The system shall process and reflect IoT sensor updates on the dashboard within 5 seconds of receipt, and support at least 500 concurrent cargo-tracking events.
Security	The system shall enforce role-based access control, encrypt data in transit and at rest, and maintain tamper-evident audit logs for all documentation transactions.
Reliability	The system shall maintain 99.5% uptime during port operating hours and gracefully degrade (cache last-known data) during connectivity loss with IoT devices.
Usability	Dashboards and forms shall be usable by non-technical port staff after no more than two hours of training, with a consistent and intuitive interface.
Scalability	The architecture shall support horizontal scaling to accommodate additional berths, terminals, and sensor networks without redesign.
Maintainability	The system shall use modular, well-documented components to allow independent updates to the AI prediction, documentation, and dashboard modules.
Availability	Core monitoring and berth-allocation services shall be available 24/7, including during scheduled maintenance windows via redundant instances.
Interoperability	The system shall expose APIs to integrate with third-party shipping-line systems, customs systems, and enterprise resource planning (ERP) platforms.

3. Actor Identification and Use Case Modeling

3.1 Primary Actors

- Port Operations Manager — allocates berths, schedules vessels, reviews dashboards and reports.

- Cargo Handling Staff — monitors and updates cargo movement, coordinates warehouse handling.
- Vessel Captain / Shipping Agent — registers vessel arrival/departure, submits shipping documentation.
- Warehouse Manager — coordinates storage, staging, and dispatch of cargo.
- Customs and Security Officer — verifies documentation, monitors cargo security and terminal access.
- System Administrator — manages users, roles, and system configuration.

3.2 Secondary Actors

- IoT / RFID / GPS Sensor Network — supplies real-time cargo and vessel location data.
- AI / Predictive Analytics Engine — processes data to forecast delays and generate insights.
- External ERP / Shipping-Line Systems — exchange scheduling and documentation data via APIs.

3.3 Use Case List

1. Register Vessel Arrival / Departure
2. Allocate Berth and Schedule Vessel
3. Monitor Cargo Movement (IoT/RFID)
4. Track Container Location (GPS)
5. Predict Cargo Handling Delays (AI)
6. Automate Logistics Documentation
7. Coordinate Warehouse Operations
8. Generate Operational Dashboard
9. Generate Performance Reports
10. Monitor Cargo Security and Access
11. Manage Users and System Configuration
12. Send Alerts and Notifications

3.4 Use Case Diagram

The diagram below shows the primary and secondary actors, the system boundary, and their associations with the major use cases of the platform.

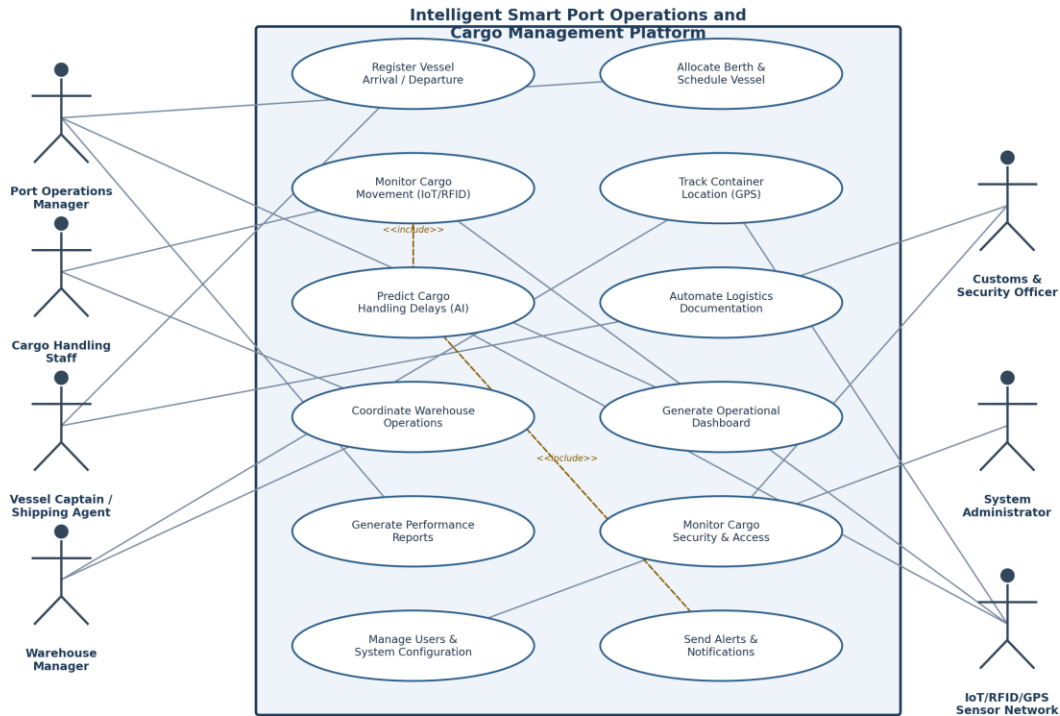


Figure 1: Use Case Diagram — Intelligent Smart Port Operations and Cargo Management Platform

3.5 Use Case Descriptions

UC-01: Allocate Berth and Schedule Vessel

Field	Description
Actors	Port Operations Manager, Vessel Captain/Shipping Agent
Preconditions	Vessel arrival request has been registered in the system.
Main Flow	1) Shipping agent submits vessel details and expected arrival time. 2) System checks berth availability against vessel size and cargo type. 3) Port Operations Manager reviews suggested allocation and confirms or overrides it. 4) System updates the schedule and notifies the vessel captain and terminal staff.
Postconditions	Vessel is assigned a berth and time slot; schedule is visible on the dashboard.

UC-02: Monitor Cargo Movement and Predict Delays

Field	Description
Actors	Cargo Handling Staff, IoT/RFID Sensor Network, AI Prediction Engine
Preconditions	Cargo has been tagged with RFID and sensors are active in the terminal.
Main Flow	1) IoT/RFID devices transmit cargo location and status updates. 2) System logs and displays movement on the dashboard in real time. 3) AI engine analyses movement patterns against historical data. 4) System flags cargo at risk of delayed handling and notifies relevant staff.
Postconditions	Cargo location is continuously tracked; delay predictions and alerts are generated when risk is detected.

UC-03: Automate Logistics Documentation

Field	Description
Actors	Vessel Captain/Shipping Agent, Customs and Security Officer
Preconditions	Vessel and cargo details are available in the system.
Main Flow	1) Shipping agent uploads or enters shipment details (manifest, bill of lading). 2) System validates data completeness and format. 3) System auto-generates required documents and routes them to customs for verification. 4) Customs officer approves or flags documentation for review.
Postconditions	Documentation is generated, validated, and stored with an auditable digital trail.

UC-04: Generate Operational Dashboard and Reports

Field	Description
Actors	Port Operations Manager, System Administrator
Preconditions	Operational data (cargo, vessel, berth, security) is being captured by the system.
Main Flow	1) Manager selects the dashboard or report type and time period. 2) System aggregates data from cargo tracking, scheduling, and documentation modules. 3) System renders real-time KPIs (turnaround time, throughput, delays) and exports the report on request.
Postconditions	Manager obtains an up-to-date view of port performance for decision-making.

4. Software Requirement Specification (SRS) Documentation

4.1 Introduction

Purpose

This document presents the Software Requirement Specification (SRS) for the Intelligent Smart Port Operations and Cargo Management Platform. It defines the functional and non-functional requirements, actors, use cases, data requirements, external interfaces, system features, assumptions, and constraints necessary to design and develop a system that digitises cargo tracking, berth allocation, and logistics documentation for a modern seaport. The document guides the design, development, testing, and validation teams, and serves as an agreement between the port authority (client) and the development team on the scope of the system.

Scope of this Document

This SRS covers the scenario-based requirement analysis for a cloud-connected, IoT-enabled port operations platform, capturing business context, stakeholder needs, requirement classification, actor-use case modelling, and validation of requirement quality, per the CO2 Assessment Tool guidelines.

4.2 Data Requirements

- Vessel data — vessel ID, name, type, size, expected/actual arrival and departure times, assigned berth.
- Cargo data — cargo ID/RFID tag, description, weight/volume, container number, current location, handling status.
- Scheduling data — berth allocation records, time slots, scheduling history and changes.
- Documentation data — manifests, bills of lading, customs declarations, gate passes, digital signatures/approvals.
- Sensor/IoT data — timestamped location and status readings from RFID, GPS, and IoT devices.
- User and access data — user accounts, roles, permissions, and authentication logs.
- Operational performance data — turnaround time, throughput, delay statistics, security incident logs.

Historical operational and sensor data shall be retained for a defined period (e.g., 24 months) to support AI model training, trend analysis, and audit requirements.

4.3 External Interface Requirements

User Interfaces

A responsive, role-based web dashboard for port managers, terminal staff, warehouse managers, and administrators; a simplified mobile-friendly interface for on-ground cargo handling staff and security personnel.

Hardware Interfaces

Interfaces to RFID readers, IoT sensors, and GPS tracking units installed on cargo, containers, and terminal equipment, communicating over standard IoT protocols (e.g., MQTT) to the cloud platform.

Software Interfaces

RESTful APIs for integration with shipping-line systems, customs/regulatory systems, and third-party ERP or logistics platforms; a cloud-based database and analytics service for storage and AI processing.

Communication Interfaces

Secure HTTPS/TLS communication for all web and API traffic; encrypted messaging for real-time sensor data ingestion; email/SMS/push-notification gateways for alerts.

4.4 System Features

Real-Time Cargo Visibility

Live tracking of cargo and container location and status using IoT and RFID data, displayed on operational dashboards.

Smart Berth and Schedule Optimisation

Automated recommendation and allocation of berths and vessel time slots to minimise waiting time and congestion.

AI-Driven Delay Prediction

Predictive analytics that flag cargo or vessels at risk of handling delays, enabling proactive intervention.

Automated Documentation Workflow

Digital generation, validation, and routing of shipping and customs documents, reducing manual paperwork and errors.

Warehouse Coordination

Coordinated storage allocation and dispatch scheduling to improve yard and warehouse utilisation.

Security and Access Monitoring

Tracking of cargo security status and terminal access control integrated with security personnel workflows.

Operational Reporting

Configurable performance reports and KPIs (turnaround time, throughput, delay trends) for management review.

4.5 Expected Outcomes

- Reduced cargo handling time.
- Improved logistics efficiency.
- Better vessel scheduling.

- Enhanced cargo visibility.
- Lower operational costs.
- Improved port productivity.
- Increased supply chain reliability.
- Better customer service.

5. Assumptions, Constraints, and Requirement Validation)

5.1 Assumptions

- Cargo and containers can be tagged with RFID/IoT devices prior to or upon entry into the terminal.
- Reliable internet and network connectivity is available across terminal, berth, and warehouse zones.
- Port staff have basic digital literacy and access to computing/mobile devices to use the platform.
- Historical operational data is available or can be collected to train the AI delay-prediction model.
- Shipping lines and customs authorities are willing to integrate with the platform via APIs.

5.2 Constraints

Type	Constraint
Technical	Dependent on stable IoT/RFID hardware and network infrastructure; integration limited by APIs exposed by third-party systems.
Operational	Port operations run continuously (24/7), constraining maintenance windows and requiring high availability.
Economic	Budget limits on the scale of IoT sensor deployment and cloud infrastructure usage.
Legal / Regulatory	Must comply with customs regulations, maritime data-security standards, and data-protection laws.
Time	The platform must be delivered within the project timeline agreed with the port authority, limiting the scope of the initial release.

5.3 Requirement Validation

Completeness Check

All eight objectives stated in the problem scenario (real-time cargo monitoring, berth/vessel scheduling, AI-based delay prediction, documentation automation, warehouse coordination, performance reporting, cargo security, and overall operational efficiency) are traceable to at least one functional requirement, use case, or system feature in this document, confirming stakeholder requirements have been addressed.

Consistency Check

Functional requirements, use cases, and system features were cross-checked to ensure no contradictory behaviour is specified (for example, berth allocation logic in FR-02 aligns consistently with UC-01, and delay-prediction logic in FR-04 aligns with UC-02).

Ambiguity and Redundancy Review

Requirements were reviewed to remove vague terms (e.g., “fast”, “user-friendly”) in favour of measurable criteria in the non-functional requirements table (e.g., 5-second update latency, 99.5% uptime). No duplicate functional requirements were identified; overlapping concerns (e.g., cargo tracking supporting both dashboards and delay prediction) were intentionally cross-referenced rather than restated.

Suggested Improvements

- Add explicit data-retention and disaster-recovery requirements for regulatory audits.
- Define measurable acceptance criteria/service-level agreements for each non-functional requirement.
- Include a change-management process for evolving customs and shipping-line integration requirements.

Conclusion

The Intelligent Smart Port Operations and Cargo Management Platform addresses the core inefficiencies of manual documentation, limited cargo visibility, and uncoordinated scheduling in modern seaport operations. By combining IoT-based real-time tracking, AI-driven delay prediction, and automated documentation and reporting, the proposed system is expected to significantly reduce turnaround time and operational costs while improving transparency, security, and overall supply-chain reliability for all stakeholders involved.